

Code-Layout ,Readability and Reusability

Date	30June, 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code Layout, Readability and Reusability

This document evaluates the quality of the **OptiCrop – Smart Agricultural Production Optimization Engine** source code in terms of structure, readability, maintainability, and reusability. The project is developed using **Python, Flask, and Machine Learning**. The code follows a modular design with proper naming conventions and organized project files, making it easy to understand, maintain, and extend.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No)	Remarks
1	Consistent Indentation	Uniform indentation and formatting maintained throughout the code.	Yes	Improves readability and debugging.
2	Proper File Structure	Project organized into templates, static, model, and application files.	Yes	Follows Flask project structure.
3	Meaningful Variable Names	Variables such as N, P, K, temperature, humidity, and prediction clearly indicate their purpose.	Yes	Easy to understand.
4	Function Names	Functions like <code>predict()</code> and <code>home()</code> are clearly named.	Yes	Improves maintainability.
5	Code Comments	Important sections include comments explaining prediction logic and model loading.	Yes	Helpful for future development.
6	Modular Design	Frontend, backend, machine learning model, and crop information are maintained separately.	Yes	Highly reusable.
7	No Redundant Code	Reusable functions and dictionaries reduce duplicate code.	Yes	Clean implementation.
8	Error Handling	User inputs are validated and exceptions are handled appropriately.	Yes	Prevents application errors.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Used	Reusability Level
1	Flask Web Application	Flask, HTML, CSS, Bootstrap	User Interface	High
2	Crop Recommendation Model	Scikit-learn, Joblib	Prediction Engine	High

3	Crop Information Module	Python Dictionary	Result Page	High
4	Static Resources	CSS, Images	User Interface	High

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project structure following Flask architecture.
Readability	5	Clear variable names, proper formatting, and simple code organization.
Reusability	5	Modular components allow easy maintenance and future enhancements.
Documentation / Comments	5	Important code sections are documented for better understanding.
Overall Score	5 / 5	Excellent code quality with good structure, readability, and maintainability.

8	ErrorHandling	Invaliduserinputsand predictionerrorsarehandled using exception handling.	Yes	Prevents application crashes.
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ReusableComponents/Modules:

S.No	Component/Module Name	Language/ Technology	Where Used	ReusabilityLevel (High/Medium/ Low)
1	FlaskWebApplication	Flask,HTML, CSS	User Interface	High
2	FeatureEncoding Module	Scikit-learn	Dataset Preparation	High
3	MachineLearning Model	RandomForest/ Logistic Regression	Prediction System	High

OverallCodeQualityAssessment:

Aspect	Rating(1-5)	Comments
CodeLayout&Structure	5	Well-organizedprojectdirectoryfollowing Flask architecture.
Readability	5	Meaningfulvariablenames,proper formatting,andcommentsimprove readability.
Reusability	5	Modular functions and reusable ML componentsallowfutureenhancements.
Documentation/ Comments	5	Adequatecommentsexplainimportant sections of the code.
OverallScore	5/5	Excellentcodequalitywith proper structure,readability,andmaintainability.